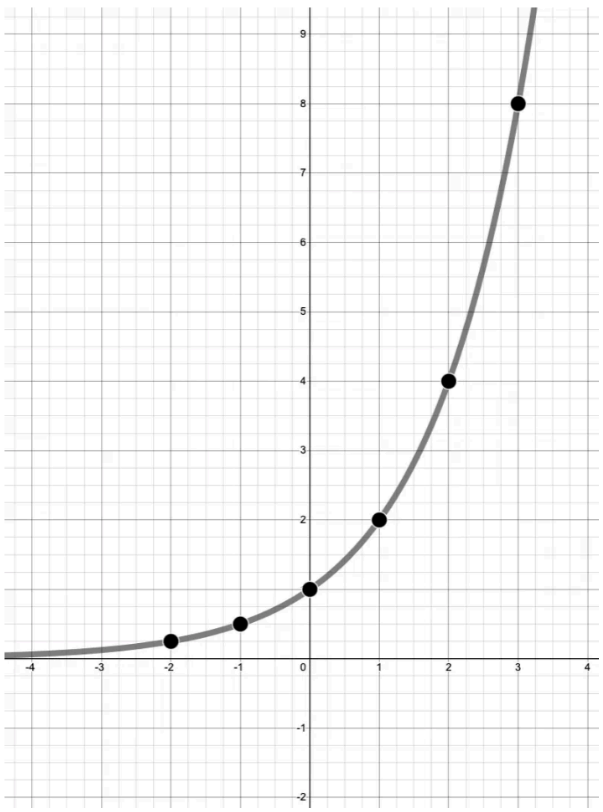
(notes) 6.2 Exponential graphs

Exponential form: $f(x) = ab^x$

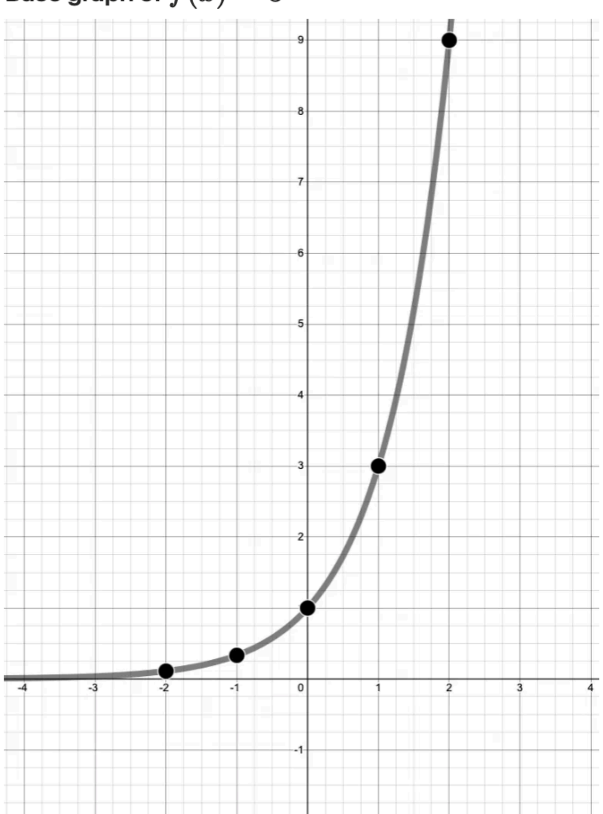
Base graph of $f(x) = 2^x$



x	$y = 2^x$
-2	
-1	
0	
1	
2	
3	

horizontal asymptote: $y =$______

Base graph of $f(x) = 3^x$



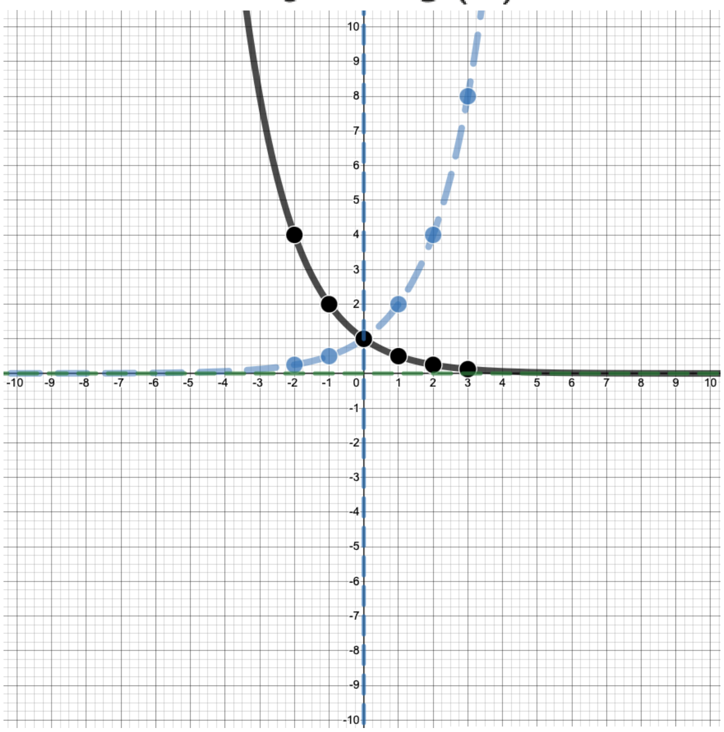
x	$y = 3^x$
-2	
-1	
0	
1	
2	
3	

horizontal asymptote: $y =$______

(notes) 6.2 Exponential graphs

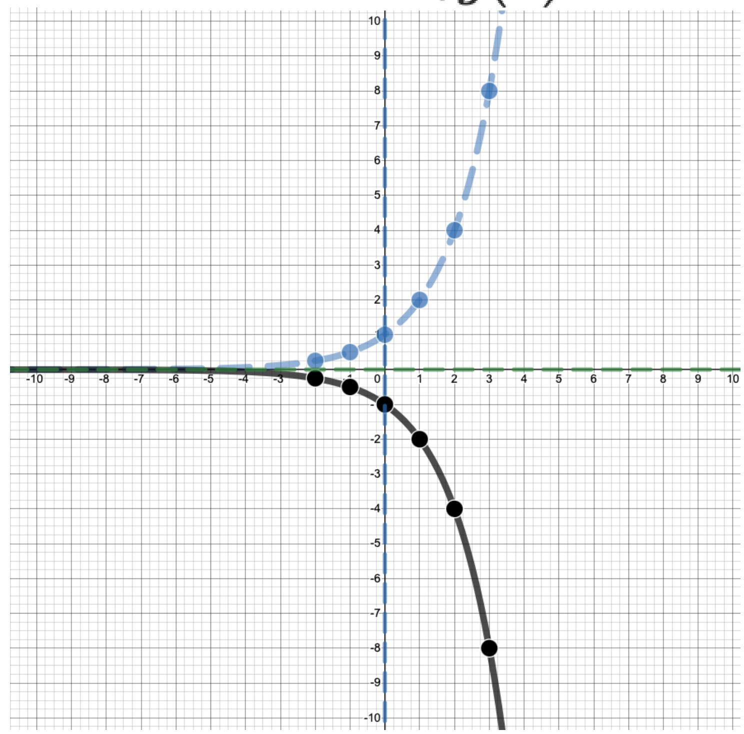
Base function $f(x) = 2^x$

Reflect over y axis, $g(x) = 2^{-x}$



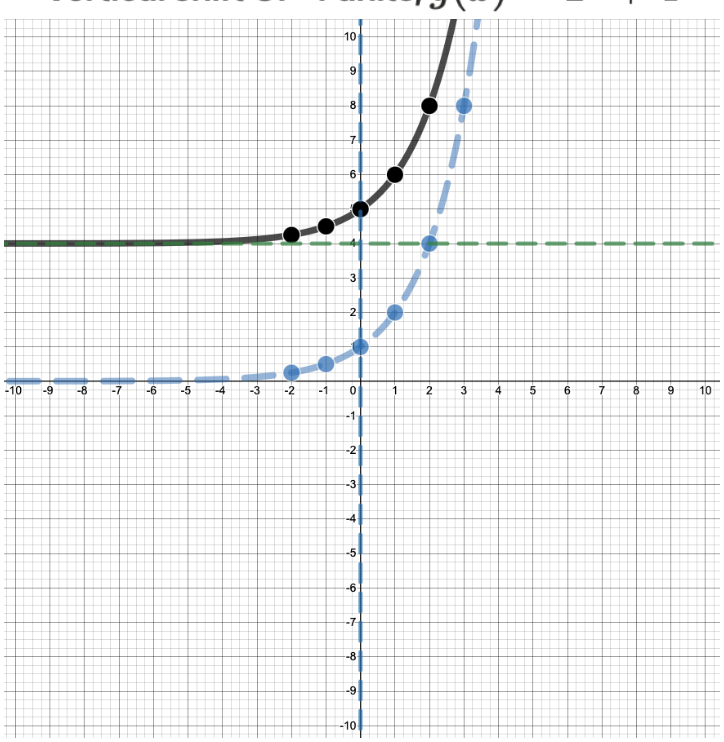
horizontal asymptote: $y =$______

Reflect over x axis, $g(x) = -2^x$



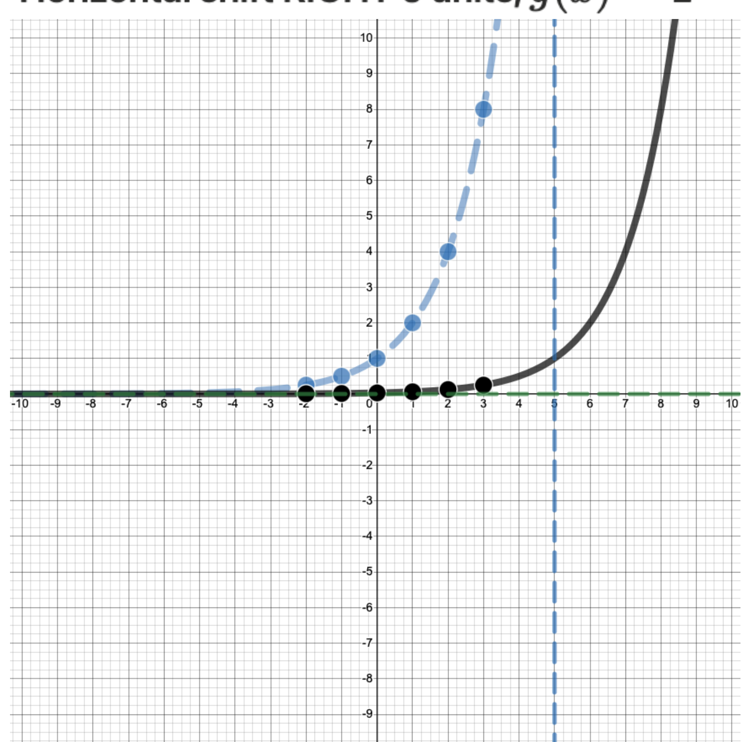
horizontal asymptote: $y =$______

Vertical shift UP 4 units, $g(x) = 2^x + 4$



horizontal asymptote: $y =$______

Horizontal shift RIGHT 5 units, $g(x) = 2^{x-5}$



horizontal asymptote: $y =$______

Graph the base function $f(x) = 2^x$

Then, plot the transformed function $g(x) = \left(\frac{1}{2}\right)^x$

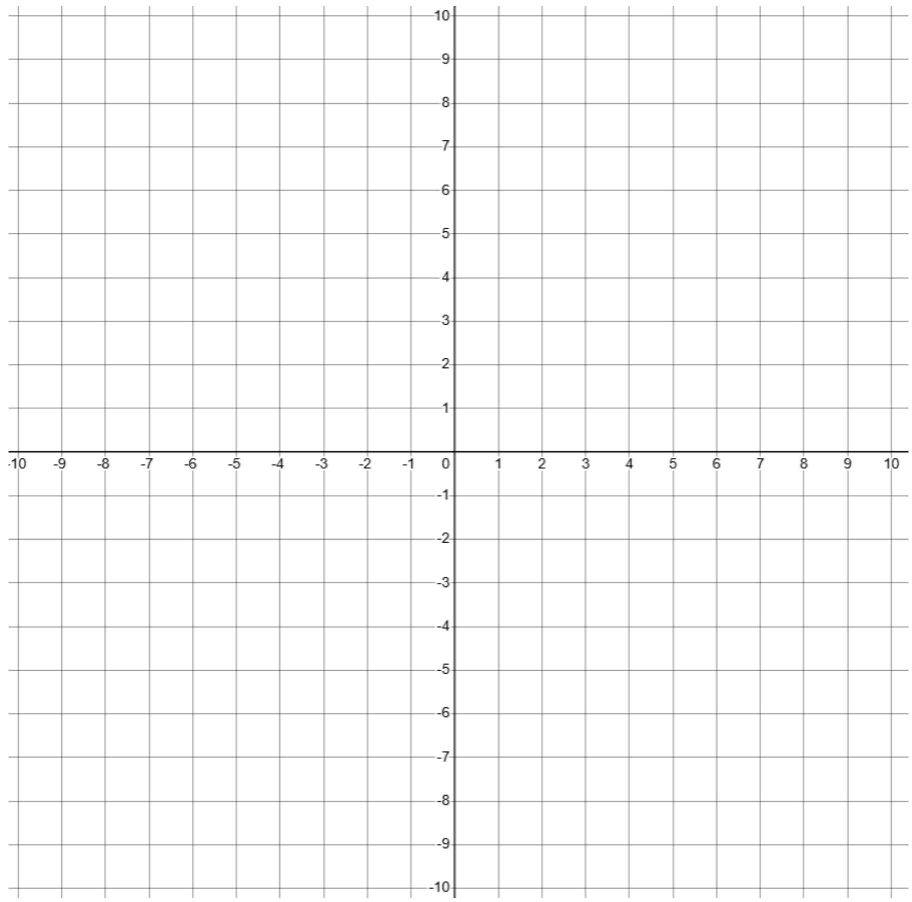
Base function: f(x) =

b=

Transformed function

a=

b=



x	$g(x) = \left(\frac{1}{2}\right)^x$
-2	
-1	
0	
1	
2	
3	

Domain (all possible x vals of function): ________________

Range (all possible y vals of function): ________________

Asymptote? vertical: x = ___ / horizontal: y = ____ / none

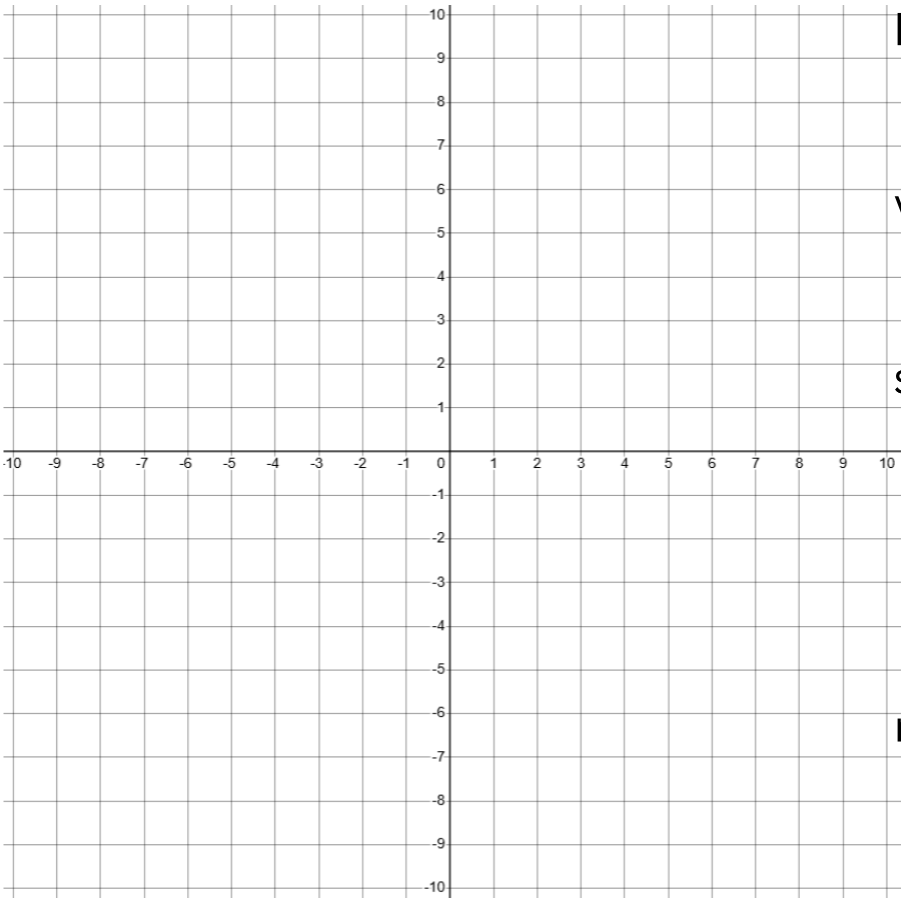
Compare the graph of the transformed function g(x) to the reflection over the x-axis. Compare the function equation of g(x) and the equation for the reflected graph.

Graph the base function $f(x) = 2^x$

Plot the transformed function $g(x) = 2^{x-3} - 7$

Base function: f(x) =

b=



horizontal shift: g(x) = f(**x +/- #**)

L / R ________ units

vertical shift: g(x) = f(x) **+/- #**

up / down ________ units

scaling factor: g(x) = **a** * f(x)

a=

- stretch, |a| is larger than 1
- compress, |a| is smaller than 1
- no scaling

reflection: g(x) = **a** * f(x)

- no , a is positive
- yes, a is negative

Domain (all possible x vals of function): ________________

Range (all possible y vals of function): ________________

Asymptote? vertical: x = ___ / horizontal: y = ____ / none

Tip when graphing:

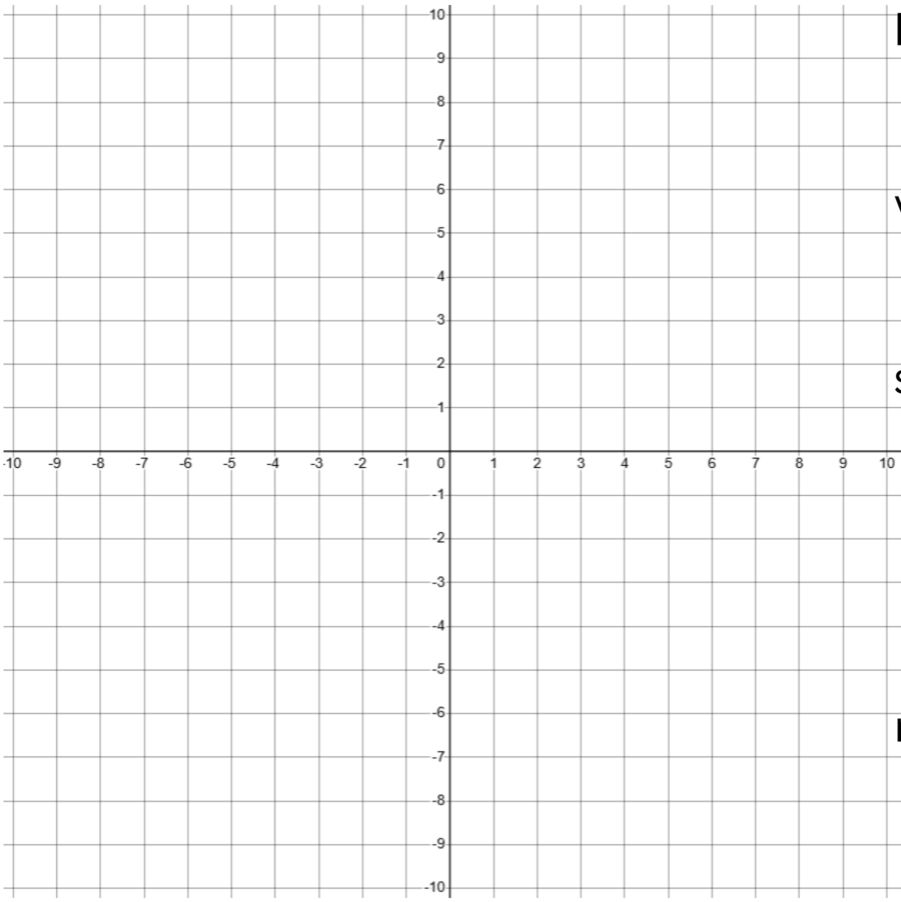
1. Identify the base graph. Plot the base graph using reference points.
2. Translation: shift your axis first. Then redraw the base function centered at your new origin. Plot the reference points, then sketch the graph.
3. Scale: stretch or compress. Use the reference points to get the correct shape.
4. Multiply the y-val of the reference point by the scaling factor.
5. Reflect

Graph the base function $f(x) = 2^x$

Plot the transformed function $g(x) = -3 * 2^x$

Base function: f(x) =

b=



horizontal shift: g(x) = f(**x +/- #**)

L / R ________ units

vertical shift: g(x) = f(x) **+/- #**

up / down ________ units

scaling factor: g(x) = **a** * f(x)

a=

- stretch, |a| is larger than 1
- compress, |a| is smaller than 1
- no scaling

reflection: g(x) = **a** * f(x)

- no , a is positive
- yes, a is negative

Domain (all possible x vals of function): ________________

Range (all possible y vals of function): ________________

Asymptote? vertical: x = ___ / horizontal: y = ____ / none

Tip when graphing:

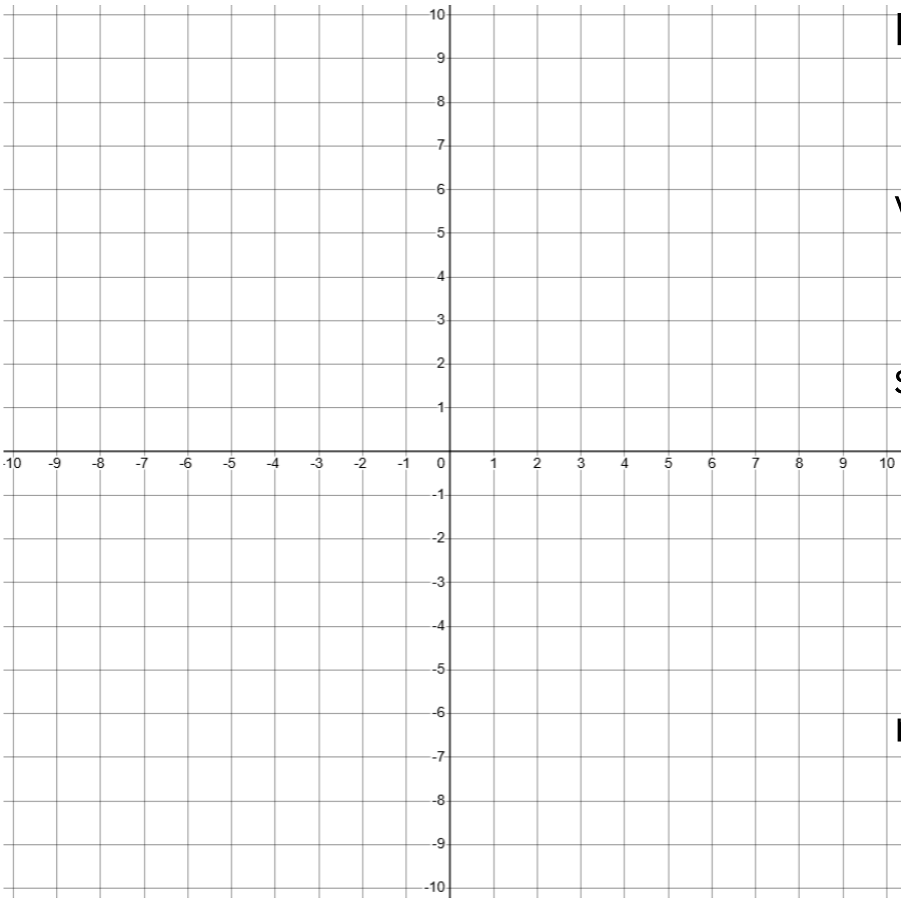
1. Identify the base graph. Plot the base graph using reference points.
2. Translation: shift your axis first. Then redraw the base function centered at your new origin. Plot the reference points, then sketch the graph.
3. Scale: stretch or compress. Use the reference points to get the correct shape.
4. Multiply the y-val of the reference point by the scaling factor.
5. Reflect

Graph the base function $f(x) = 3^x$

Plot the transformed function $g(x) = 3^{-(x-4)}$

Base function: f(x) =

b=



horizontal shift: g(x) = f(**x +/- #**)

L / R ________ units

vertical shift: g(x) = f(x) **+/- #**

up / down ________ units

scaling factor: g(x) = **a** * f(x)

a=

- stretch, |a| is larger than 1
- compress, |a| is smaller than 1
- no scaling

reflection: g(x) = **a** * f(x)

- no , a is positive
- yes, a is negative

Domain (all possible x vals of function): ________________

Range (all possible y vals of function): ________________

Asymptote? vertical: x = ___ / horizontal: y = ____ / none

Tip when graphing:

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4. Multiply the y-val of the reference point by the scaling factor.
5. Reflect